PHISHING AWARENESS ANALYSIS

Internship Project Report

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Project Title: Phishing Awareness Analysis

Organization: DecodeLabs

DECLARATION

I hereby declare that the project titled "Phishing Awareness Analysis" submitted as part of my Cybersecurity Internship is an original work carried out by me. The project has been developed using my own efforts under the guidance and resources provided during the internship period.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my internship organization for providing me with the opportunity to work on the project "Phishing Awareness Analysis". This project helped me gain practical knowledge of cybersecurity concepts and understand the importance of identifying phishing attacks and online threats.

I am thankful to my mentors and guides for their valuable support, guidance, and encouragement throughout the project. Their suggestions and feedback helped me successfully complete this work.

I would also like to thank everyone who directly or indirectly contributed to the completion of this project and supported my learning during the internship period.

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1.Abstract

Phishing attacks are one of the most common cybersecurity threats used by attackers to steal sensitive information such as usernames, passwords, and banking details. This project focuses on creating a simple Python-based Phishing Awareness Analysis tool that identifies suspicious keywords and common phishing indicators in messages or emails. The project helps users understand how phishing attacks work and promotes cybersecurity awareness.

2.Introduction

Phishing is a cyberattack technique where attackers impersonate trusted organizations to trick users into revealing confidential information. Many users become victims because they are unable to recognize phishing attempts. This project demonstrates a basic approach to phishing detection by analyzing message content and identifying suspicious patterns commonly used in phishing attacks.

3.Objective of the Project

- To understand phishing attacks and their impact.
- To identify suspicious keywords in messages.
- To increase cybersecurity awareness.
- To demonstrate a simple phishing detection mechanism using Python.
- To help users recognize common phishing indicators.

4.Problem Statement

Many internet users receive fraudulent emails and messages that appear legitimate. These messages often contain urgent requests, fake links, or attempts to collect sensitive information. The objective of this project is to analyze such messages and detect potential phishing attempts based on predefined indicators.

5.Project Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet Connection (Optional)

Software Requirements

- Python 3.x
- Visual Studio Code / PyCharm
- Windows/Linux Operating System

6.System Design

Input:

- Email or Message Content

Processing:

- Scan for suspicious phishing keywords
- Analyze message characteristics
- Identify phishing indicators

Output:

- List of suspicious keywords detected
- Security warning and analysis
- Phishing detection result

7.Methodology

1. Create a list of commonly used phishing keywords.
2. Accept or define a sample email/message.
3. Compare message content with phishing keywords.
4. Detect suspicious words and phrases.
5. Display identified threats and analysis.
6. Generate a final phishing detection result.

8.Implementation

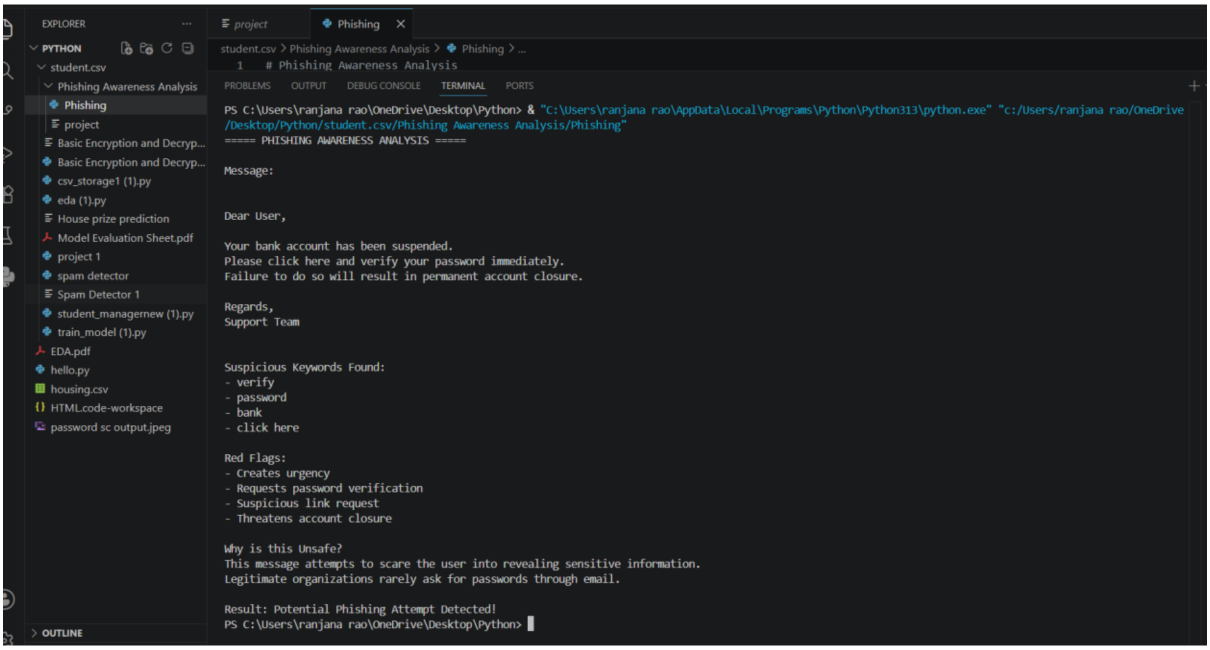
The project is implemented using Python programming language. A predefined list of phishing-related keywords is used for comparison. When the program finds matching keywords in the message, it flags them as suspicious and provides security recommendations.

9.Source Code

```
# Phishing Awareness Analysis


phishing_keywords = [

  "urgent", "verify", "password", "bank",

  "login", "click here", "account suspended",

  "update", "winner", "claim"

]


sample_message = """

Dear User,


Your bank account has been suspended.

Please click here and verify your password immediately.

Failure to do so will result in permanent account closure.


Regards,

Support Team

"""


print("===== PHISHING AWARENESS ANALYSIS =====")

print("\nMessage:\n")

print(sample_message)
```

```python
found_keywords = []

for keyword in phishing_keywords:
    if keyword.lower() in sample_message.lower():
        found_keywords.append(keyword)

print("\nSuspicious Keywords Found:")
for word in found_keywords:
    print("-", word)

print("\nRed Flags:")
print("- Creates urgency")
print("- Requests password verification")
print("- Suspicious link request")
print("- Threatens account closure")

print("\nWhy is this Unsafe?")
print("This message attempts to scare the user into revealing sensitive information.")
print("Legitimate organizations rarely ask for passwords through email.")

if found_keywords:
    print("\nResult: Potential Phishing Attempt Detected!")
else:
    print("\nResult: No major phishing indicators found.")
```

10.Output Screenshots



11.Results and Analysis

The program successfully identifies suspicious phishing keywords such as "verify", "password", "bank", and "click here". Based on the detected indicators, the tool warns the user about a potential phishing attempt. The project demonstrates how simple text analysis can improve cybersecurity awareness.

12.Applications

- Cybersecurity Awareness Training

- Educational Demonstrations

- Student Projects

- Basic Email Security Analysis

- Security Awareness Workshops

13.Limitations

- Uses only keyword-based detection.

- Cannot detect advanced phishing techniques.

- Does not analyze URLs or attachments.

- Limited to predefined phishing indicators.

14.Future Scope

- Integration with Machine Learning algorithms.

- Real-time email scanning.

- URL reputation checking.

- Advanced phishing detection techniques.

- Graphical User Interface (GUI) development.

15.Conclusion

The Phishing Awareness Analysis project successfully demonstrates a simple method of detecting phishing attempts through keyword analysis. The project highlights common phishing indicators and helps users understand cyber threats. It serves as a useful educational tool for learning basic cybersecurity concepts and promoting safe online practices.

16.References

1. Python Official Documentation

2. OWASP Security Guidelines

3. Cybersecurity Awareness Resources

4. Information Security Learning Materials

5. Online Phishing Awareness Articles